DSS 5202 Sustainable Systems Analysis

Term Paper Topic Proposal

Group No: 29

Members:

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Title: The Thirsty Cloud: Meeting the Water Needs of Singapore's Data-Centre Growth

Description:

Singapore is expanding its data-centre capacity to support growing digital and AI demand. In August 2026, 200 MW of new data-centre capacity was provisionally awarded under Singapore's second Data Centre Call for Application. As data-centre capacity grows, more water may be needed for cooling, depending on the cooling technologies used.

This creates a wider sustainability issue for Singapore. As Singapore's water demand grows, NEWater and desalination are important weather-resilient sources of supply, although both require energy for treatment and therefore have financial and carbon costs.

This project will study how Singapore can address the additional cooling-water needs from new data-centre capacity between 2027 and 2035. We will compare two broad approaches. The first is to meet the additional water requirement through water-supply options such as desalination and NEWater. The second is to reduce the amount of additional water that needs to be supplied through more water-efficient cooling and measures such as on-site water reuse.

The aim is not to determine which cooling technology is technically the best. Instead, we want to understand the trade-offs between **supplying more water and reducing the amount of water that needs to be supplied**, and how these choices affect cost, carbon emissions, water security and other sustainability concerns.

Stakeholders:

- **PUB**, which manages Singapore's water supply and water-efficiency policies.
- **IMDA and EDB**, which oversee data-centre growth and related sustainability requirements.
- **Data-centre operators**, which make decisions about cooling systems, water use and on-site water reuse.
- **EMA**, because both water treatment and data-centre cooling depend on electricity.
- **Other water users**, including households and industries that share Singapore's water resources.
- **Environmental stakeholders**, particularly in relation to the energy use and environmental impacts of additional water production.

Objectives:

- Estimate the additional annual cooling-water demand from Singapore's data-centre growth by 2035 under different cooling-efficiency assumptions.
- Compare desalination, NEWater/reclaimed water, and selected demand-side water-efficiency measures in terms of life-cycle cost and greenhouse-gas emissions, where suitable data are available.
- Assess the options by considering cost and carbon emissions together with broader factors such as water security and environmental impacts.
- Identify which assumptions have the greatest effect on the results and provide practical recommendations for PUB and IMDA.

Sustainability and other issues to be addressed:

The project focuses on the link between **water, energy and data-centre growth**. Producing additional water through desalination or NEWater requires electricity, which creates costs and carbon emissions. On the other hand, reducing cooling-water use may require changes in cooling technology or investment in water-reuse systems.

The project will therefore consider the trade-off between increasing water supply and improving water-use efficiency. It will also consider water security, carbon emissions, cost and environmental impacts.

Questions to be answered:

The project will answer the following questions:

1. How much additional cooling water could Singapore's new data-centre capacity require by 2035?
2. How much can the need for additional water supply be reduced through more water-efficient cooling and on-site water reuse?
3. What are the life-cycle cost and carbon impacts of the main options for addressing the additional water requirement?
4. How do the alternatives compare when water security, cost, carbon emissions and environmental impacts are considered together?
5. How sensitive are the results to uncertain assumptions such as cooling-water efficiency, desalination energy use and future electricity-grid emissions?

Proposed approach:

1. **Define the study boundary.** The study will focus on the additional cooling-water demand associated with Singapore's data-centre growth up to 2035. The exact comparison basis and functional unit will be defined after reviewing the available data and relevant literature.
2. **Estimate the additional cooling-water demand.** A simple demand model will be developed using data-centre capacity, utilisation and water-use effectiveness (WUE). Different cooling-efficiency assumptions will be used to estimate a reasonable range of additional annual water demand by 2035.
3. **Compare the main options.** We will assess three ways of addressing the additional water requirement: desalination, NEWater/reclaimed water, and demand-side measures such as more water-efficient cooling and on-site water reuse. Life Cycle Costing (LCC) will be used to compare costs, while a simplified Life Cycle Assessment (LCA) will compare energy use and greenhouse-gas emissions using published data.
4. **Consider broader decision factors.** Analytic Hierarchy Process (AHP) will be used to assess the options by considering the LCC and LCA results together with broader factors such as water security and environmental impacts. Published information on stakeholder concerns will help inform the weighting scenarios, rather than being treated as exact stakeholder preferences.
5. **Test uncertain assumptions.** Sensitivity analysis will be used to test how changes in key assumptions affect the results. These may include assumptions related to data-centre water use, water-treatment energy use and future grid emissions. The final parameters and ranges will be selected based on the data available and the findings from the main analysis. A low-carbon-grid case will also be considered to examine how future grid decarbonisation could affect the results. Nuclear power will not be modelled separately but may be discussed as one possible contributor to a future lower-carbon electricity system.
6. **Combine the findings and develop recommendations.** The results from the demand model, LCA, LCC, AHP and sensitivity analysis will be brought together to identify the main trade-offs and develop practical recommendations for PUB and IMDA on water planning and data-centre water-efficiency requirements.

Systems analysis or assessment methodologies to be used:

The project will use three main methods: simplified Life Cycle Assessment (LCA), Life Cycle Costing (LCC), and Analytic Hierarchy Process (AHP). Scenario and sensitivity analysis will be used across the models.

Sources of information or data:

- **PUB:** Information on NEWater, desalination, water demand, water-efficiency measures and tariffs.
- **IMDA and EDB:** Data and information on data-centre growth, planned capacity and the Green Data Centre Roadmap.
- **EMA:** Electricity data, electricity prices and Singapore's grid emission factor.
- **SS 726:2026 and industry sources:** Information on data-centre cooling and water-efficiency practices.
- **Data-centre operators:** Sustainability reports and published information on water use, WUE and cooling performance.
- **Academic literature:** Peer-reviewed studies on data-centre water use, desalination, water reclamation, LCA and LCC.
- **Data gaps:** Where site-level data are not publicly available, published ranges will be used and tested through sensitivity analysis.

Expected final deliverables:

A term paper with documented assumptions and references, supporting calculations or models for the main analyses, a poster and the video presentation required by the module.

Any other information:

The scope has been kept deliberately focused for a three-person team. The aim is to apply a smaller set of methods carefully and connect them into one clear decision problem rather than use many methods with limited depth.

References

Economic Development Board & Infocomm Media Development Authority. (2026). Four data centre proposals selected under the second Data Centre - Call for Application.

Enterprise Singapore & Infocomm Media Development Authority. (2026). Singapore launches world's first liquid cooling standard for data centres in tropical climates (SS 726:2026).

Energy Market Authority. (2025). Singapore Energy Statistics 2025.

Li, P., Yang, J., Islam, M. A., & Ren, S. (2023). Making AI less 'thirsty': Uncovering and addressing the secret water footprint of AI models.

Ministry of Sustainability and the Environment. (2023). PUB's decarbonisation strategy.

Mytton, D. (2021). Data centre water consumption. npj Clean Water, 4, 11.

PUB. (n.d.). Desalinated water; NEWater; sustainability and water-efficiency information.

Saaty, T. L. (1980). The analytic hierarchy process. McGraw-Hill.